

Rodrigo Sobral

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SUMMARY

DevOps Engineer with a strong foundation in Software Engineering and a specialization in Cybersecurity. Experienced in building and operating large-scale platforms in highly demanding environments such as CERN, holding both the CKA and CKS certifications. Career path spanning backend development, microservices architecture, and platform engineering - now focused on secure, resilient, and observable infrastructure at scale.

EXPERTISE

Cybersecurity: Kubernetes security hardening, RBAC, network policies, admission controllers; secure infrastructure design, defense-in-depth, least privilege, IaC security; risk assessment & management; input validation, compliance and security best practices; penetration testing, MITRE ATT&CK, Kill Chain, malware analysis

DevOps & Platform Engineering: Kubernetes administration and operations; GitOps with ArgoCD and Helm; Infrastructure as Code with Terraform; container orchestration with Docker; CI/CD pipeline design and automation; cloud infrastructure on OpenStack

Software Engineering: Backend development in Python; REST API design and integration; microservices architecture; PostgreSQL, Redis, AWS S3, and SAP integrations; legacy code refactoring; software development lifecycle (Agile/Scrum, Kanban, Waterfall)

Monitoring & Observability: Observability pipelines with Grafana and Prometheus; real-time threat detection with Falco; workflow security auditing; incident response, anomaly detection, and threat correlation; user behavioral analysis

EXPERIENCE

CERN · DevOps Engineer

Jan 2024 – Present · Geneva, Switzerland

- Led a cross-departmental partnership project between the **IT** and **ATS** departments at CERN (one of the world's largest and most complex scientific computing environments) to develop, deploy, and operate an internal variant of **SWAN** (Service for Web-based ANALysis), a large-scale Jupyter-based data analysis platform, bringing new features like **Custom Software Environments** and serving tens of users across the ATS department
- Presented the project at multiple high-visibility internal events including the **ITCSC**, driving awareness and adoption across CERN
- Managed main instance of SWAN, with a total of **~400 users** daily and two production clusters composed of **54 nodes** in total, contributing with new features and improvements, like the implementation of **uv**, optimising enviroment creation runtime by **~77%**
- Implemented real-time **threat detection** tools like Falco, to monitor suspicious activity across user's sessions on **Kubernetes** clusters, and zizmor, to alert on and bad practices across all workflows
- Maintained **observability pipelines** (Grafana, Prometheus) to monitor platform health and address incidents, by providing **user support** and issue resolution via ServiceNow and JIRA
- Supervised interns and new team members, providing **mentorship** and guidance on the platform's architecture, best practices, and operational procedures

UCNext · Backend Developer

Oct 2022 – Oct 2023 · Coimbra, Portugal

- Delivered systems serving **~34,000** students, teachers, researchers and staff across the **University of Coimbra's** digital ecosystem
- Led the backend development of several production platforms: **UCCompetitions**, **UCApply**, **MyUC**, and internal tools including **UCAnalytics** (subject for **research paper**), **UCDigitalSignature**, and **UCID** (building access control)
- Designed and implemented **REST APIs** integrated with **PostgreSQL**, **Redis** caching layers, **AWS S3**, and **SAP** systems

Superprof · Private Tutor

Jun 2022 – Jul 2023 · Remote

- Delivered remote one-on-one tutoring to computer science students, covering core engineering fundamentals across multiple disciplines. Helping them to strengthen their understanding of key concepts and prepare for exams and projects, including: algorithms, networking, databases, data structures, operating systems.

Project Box · Analyst Developer

Apr 2022 – Sep 2022 · Viseu, Portugal

- Led the backend development of **UCNotes**, a new internal note taking tool, inserted across a **distributed microservices** system
- Performed targeted **refactoring of legacy code** to improve maintainability, performance, and alignment with modern backend standards

Critical Software · Summer Intern

Jul 2020 – Aug 2020 · Coimbra, Portugal

- First professional exposure to the software industry, working on a prototype through a full software development lifecycle using both **Waterfall** and **Agile methodologies** (Scrum/Kanban).

EDUCATION

2021 – 2023 · **M.Sc. Cybersecurity** - University of Coimbra · Thesis: **Analysis of user behavioral patterns in cybersecurity**

2019 – 2024 · CTF Player (Offensive Security Training) - **TryHackMe**

2018 – 2021 · **B.Sc. Computer Science** - University of Coimbra

CERTIFICATIONS

CKS: Certified Kubernetes Security Specialist · **CKA: Certified Kubernetes Administrator** · **Junior Cybersecurity Analyst Career Path** · **Ethical Hacker** · **Others**

LANGUAGES

Portuguese - Native | English - Fluent | Spanish - Intermediate | French - Basic (A2)